

Finger Painting in VR: Multi-Dynamic Gestural Input for VR Painting

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Abstract

This study explores the multi-dynamic gestural interaction system using VR headsets and wearable devices, for enhancing VR painting with pressure control, gesture recognition, and haptic feedback. Testing demonstrated novel visual expressions through hand gestures, suggesting the potential for more nuanced and dynamic interactions in VR art and beyond.

CCS Concepts

• **Human-centered computing**; • **Human computer interaction (HCI)**; • **Interaction devices**; • **Haptic devices**;

Keywords

Virtual Reality, Art, Haptic device, Gestural recognition

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1 INTRODUCTION

As an exploration of using virtual reality (VR) headsets for creative practices, this paper discusses the potential of a multi-dynamic interaction system for artistic expressions using wearable devices. We investigate VR as an embodied medium that tracks and modulates bodily movements with attention to hand gestures as an important element for dynamic visual expressions.

We recognized that the current VR interactions for generating tracings lack delicate and nuanced hand controls that are often integral to painting and drawing with the physical medium in art.

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Inspired by the multifaceted dynamism of the Chinese ink paintings, we explored capturing finger gestures for tracing and size control and hand forces with a pressure sensor for opacity control of VR brushstrokes. Moreover, we implemented two motors for haptic feedback at the tips of the index and ring fingers.

2 ATTRIBUTES AND LIMITATIONS OF VR DRAWING

VR controllers are capable of tracking bodily tracings in real-time but does not facilitate more nuanced hand gestures used in tangible mediums. Reviewed studies that utilised immersive VR indicated limitations due to the mandatory use of motion capture for the hands, facilitated by controllers or other techniques, often resulting in users experiencing a disembodied sensation [Hadjipanayi et al., 2023].

In Old English, the word "drawing" are *dragan*, meaning "to draw, drag, or pull."; where in Sanskrit, "*dhrājas*" means "gliding course or motion" [Mustaqim, 2018]. In drawing, the body adapts and reacts to the materials dynamically, deriving specific gestural techniques based on different materials. Similarly, the process of embodying VR as a drawing medium is influenced by bodily sensations and intuitions. VR as a creative medium highlights spontaneous and subtle bodily movements. Without tangible feedbacks, VR drawing is more susceptible to unconscious movements like walking and leaning. The absence of substrate and friction in VR drawing introduces uncertainty in control. How can we address the sense of "immateriality" in VR drawing?

Recognising the potential of bodily expressions in VR, we differentiate VR painting from VR drawing. VR painting involves generating dynamic 'brushstrokes,' similar to those in wet mediums, rather than just generic tracings. This perspective led us to explore how mediums respond to the dynamic forces of the hand.

Inspired by the fluidity and diversity in ink and its intimate relationship with the body in Chinese calligraphy and painting, we reflect on the dynamic expressions that are lacking in controller-based interactions. The liquid medium coupled with soft brush tips is very susceptible to the speed, angle and forces of the motion [Self and Self, 2009]:

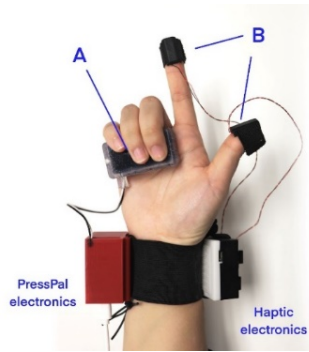


Figure 1: (a) PressPal (b) Fingertip haptic feedback device

“Almost all the strokes are made using the whole body. This unity of motion not only produces beautiful strokes but rewards the artist with a kinesthetic sensation that is very pleasing.”

3 DESIGN AND IMPLEMENTATION

Our study uses Meta Quest Pro as the operational platform, it expands on the VR painting experience by enabling dynamic control of brush width and opacity. This is achieved by tracking and calibrating the hand's gesture and forces. We designed a wearable device with a handheld pressure sensor for dynamic input(a); We also developed a haptic feedback device targeting the user's thumb and index finger to replicate the audio and tactile sensations inherent in drawing and painting with tangible mediums(b).

3.1 Gesture recognition

Using the Presence Platform Interaction SDK, we implemented a hand gesture recognition feature that allows the users to adjust the brush width by varying the distance between the thumb and index finger. Users can also perform a double pinch gesture with their left hand to execute an undo action.

3.2 PressPal

PressPal is made of a soft silicone tactile sensor that changes capacitance value when deformed by pressure. The sensor was chosen as it is that it is elastic and deformable, which is reminiscent of a flexible paintbrush tip that changes the stroke when pressed onto the paper [Strohmeier et al., 2018]. Pressure value is correlated with the opacity of the brush strokes. The more forcefully the user squeezes PressPal, the darker the strokes appear.

3.3 Haptic device

The fingertip haptic feedback device uses two piezoelectric motors, driven by a custom circuit board connected to the PC. These motors offer a wider vibration bandwidth and lower latency than conventional electromagnetic motors, while maintaining a compact size ideal for wearable haptics.

Motion-coupled haptic feedback forms [Mahmoudinezhad et al., 2022] the basis of our haptic rendering algorithm, where the user's translating motion triggers vibration pulses at certain intervals. Additionally, the haptic signals were rendered according to the two



Figure 2: VR painting of a bird resting on a tree using multi-dynamic gestural input

dynamics of user input: where the haptic intensity is associated with brush size and vibration pitch is associated with opacity.

4 DEMO AND DISCUSSION

We tested the interactive system by painting playful figures in VR. Although it requires practice to be accustomed to but the experience was enjoyable and distinctive from painting with VR controllers. The user can manipulate the expressions of the brushstrokes in one fluid motion. The changing width and opacity of the brushstrokes enhance the 'liveliness' of the artwork and show resemblance to characteristics of ink paintings. The demonstrations suggest an emerging art practice between ink painting and VR.

5 CONCLUSION

In designing the multi-dynamic gestural interaction system, we aimed to foreground the rich expressions of human body in digital art practices. The study demonstrates the positive potential of using wearable devices to enhance bodily expressions in VR. An improvement we could work on is to correlate the dynamic inputs with other visual expressions such as colour and texture. These features has been useful in digital illustration and 3D modelling practices. The dynamic expressions of the hand can even be used for playing musical VR instruments.

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